Civics Group	Index Number	Name (use BLOCK LETTERS)

H2

ST. ANDREW'S JUNIOR COLLEGE
2022 JC2 PRELIMINARY EXAMINATIONS

H2 BIOLOGY **9744/01**

Paper 1: Multiple Choice

Friday 16th September 2022 1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **30** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice Optical answer sheet.

INFORMATION TO CANDIDATES

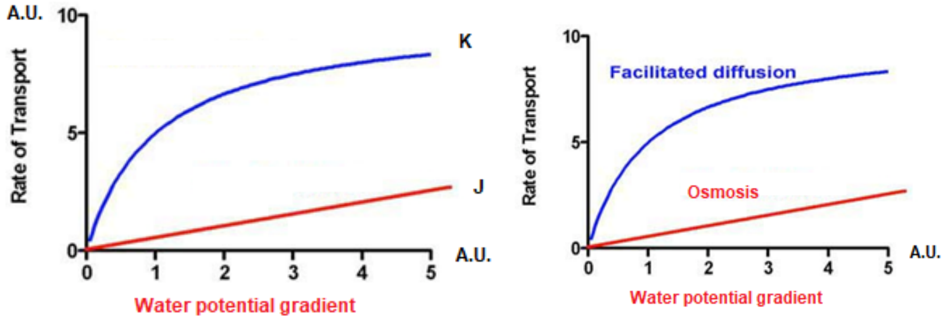
Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit both question paper and multiple choice answer sheet.

1	Which group of eukaryotic cell structures all contain nucleic acids?	
	A	cytoplasm, Golgi bodies, mitochondria, nuclei
	B	centrioles, chloroplasts, mitochondria, ribosomes
	C	centrioles, mitochondria, nuclei, ribosomes
	D	chloroplasts, mitochondria, cytoplasm, ribosomes

2	Which properties of phospholipids explain why single layers of phospholipids added to water immediately form bilayers? 1 The hydrophobic fatty acid chains repel water molecules so the tails pack together. 2 The non-polar fatty acid chains are attracted to each other by hydrophobic interactions. 3 Charged phosphate groups form hydrogen bond with water. 4 Phospholipids are insoluble in water	
	A	1, 2 and 3
	B	1 and 2 only
	C	1 and 4 only
	D	2 and 3 only

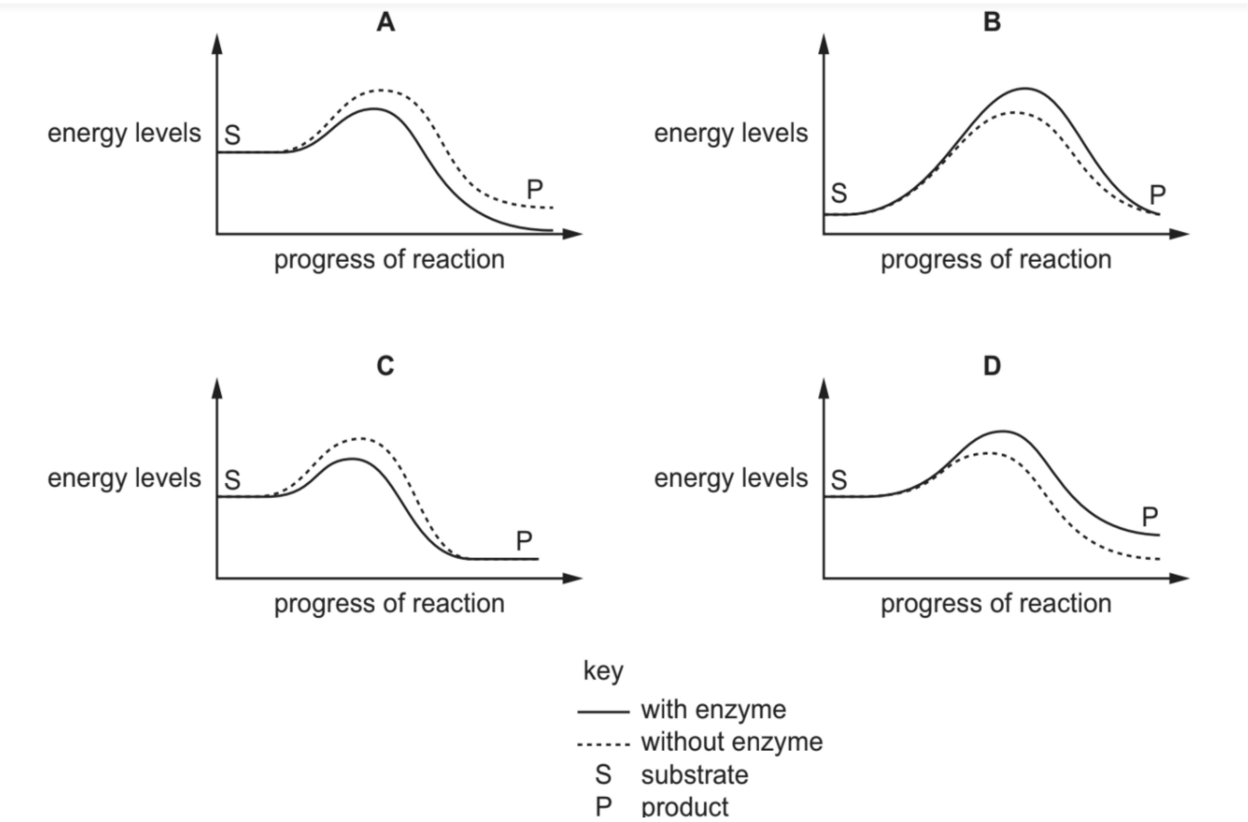
3 The graph shows rates of osmosis and facilitated diffusion of water molecules (through aquaporins) across the cell surface membrane.



Which row is correct?

		graph K	Explanation
	A	facilitated diffusion	limited by the number of aquaporins in the cell surface membrane
	B	Osmosis	allows the concentrations of water molecules inside and outside the cell to be equal in a shorter time
	C	facilitated diffusion	rate of transport approaches plateau at high water potential gradient due to dynamic equilibrium being achieved
	D	Osmosis	limited by the number of transient gaps in the cell surface membrane

4 Which graph correctly shows possible changes in energy levels as a chemical reaction progresses with or without an enzyme? **Answer C**



5 In 2016, the United Kingdom government allowed scientists to genetically modify cells obtained from very early human embryos(immediately after fertilization till 8 cells stage).

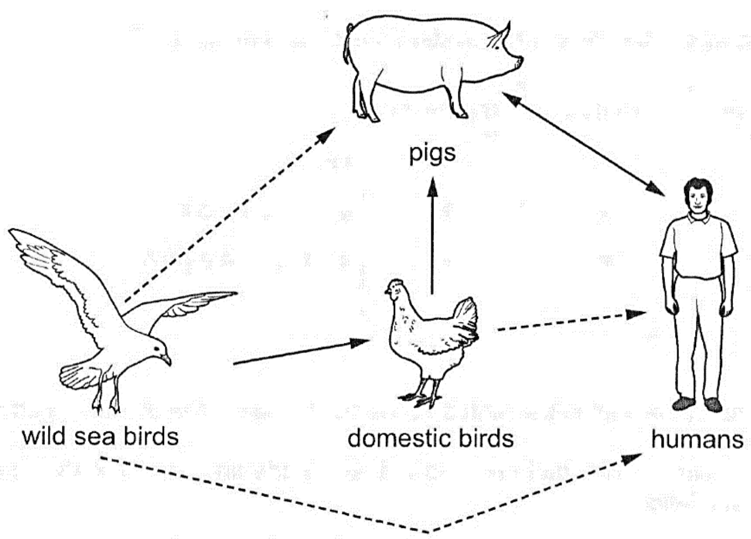
Which statement correctly describes an ethical concern about this research?

A	Genetically modified iPSCs could give rise to genetically modified individuals.
B	Genetically modified multipotent stem cells could give rise to genetically modified individuals.
C	Genetically modified pluripotent stem cells could give rise to genetically modified individuals.
D	Genetically modified totipotent stem cells could give rise to genetically modified individuals.

6 The sub-types of the influenza A virus that infect birds, humans and pigs in one area of the world in recent times are shown in the table.

	influenza A virus sub-types present		
time period	birds	humans	pigs
1918–1957	show any one of the H1–H16 antigens combined with any one of the N1–N9 antigens	H1N1	H1N1
1958–1970		H2N2	
1971 to present day		H3N2 H1N1	H3N2 H2N3

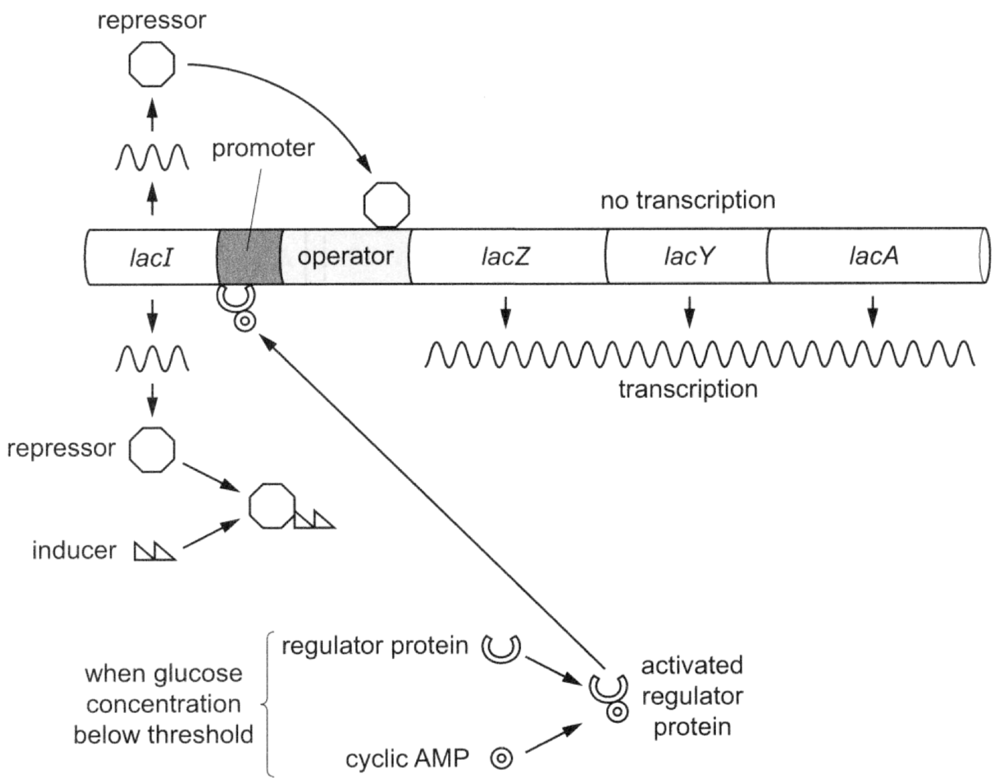
High risk of transmission of influenza A virus between species is shown by solid arrows in this diagram. A dotted arrow shows a low risk of transmission.



Which statement correctly describes a danger to human health in this area of the world?

A	Antigenic drift of human viruses such as H3N2, leading to vaccines being less effective
B	Antigenic drift within pigs, leading to emergence of H2N2 virus able to infect humans
C	Antigenic shift of bird viruses, leading to emergence of new viruses such as H7N10
D	Antigenic shift within humans, combining H2N2 from older people with H1N1 or H3N2

7 The diagram shows some of the factors affecting the expression of the lac operon.

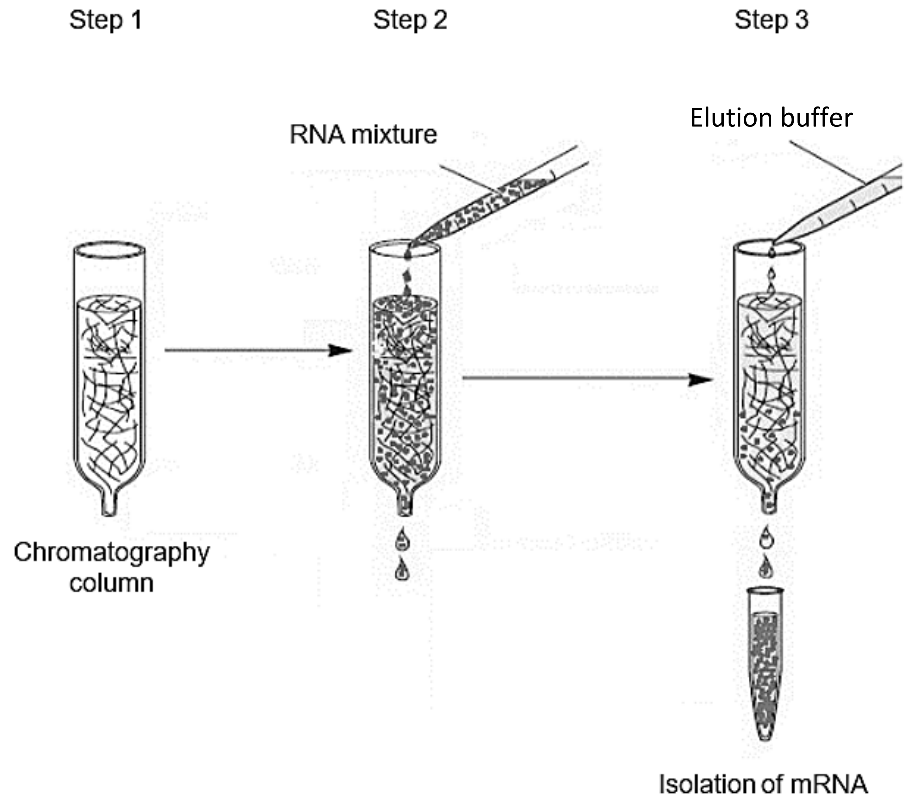


Only when the concentration of glucose is below a critical threshold can the lac operon be fully induced by the presence of lactose. As the concentration of glucose decreases, the concentration of cyclic AMP increases.

Which statement describes the role of cyclic AMP in the expression of the lac operon?

	A	It prevents the regulator protein from binding to the promoter when the glucose concentration is high.
	B	It activates the inducer to bind to the lac operator.
	C	It prevents the expression of the lac operon when the glucose concentration is high.
	D	It activates a regulator protein that helps RNA polymerase bind more effectively to the promoter.

8 The isolation of mature mRNA can be done by passing an RNA mixture obtained from homogenised tissue through a chromatography column. The procedure of the mRNA isolation is as follows:



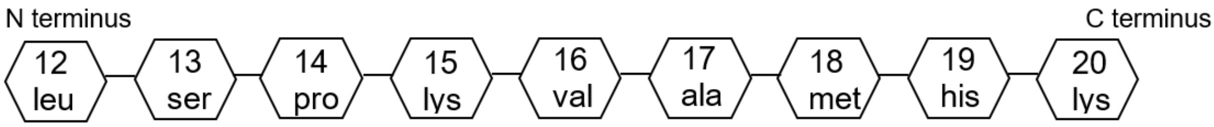
Step 1	Set up a chromatography column which contains short lengths of uracil nucleotides attached to a solid support medium.
Step 2	Add RNA mixture to the chromatography column. RNA that do not hybridise with the uracil nucleotides will pass through and leave the column.
Step 3	Add water to the chromatography column to remove and isolate the hybridised mRNA.

Which of the following statements is **not** true?

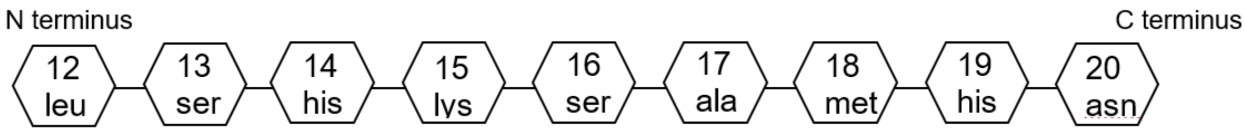
A	rRNA and tRNA molecules could pass through and leave the column in Step 2.
B	Complementary base pairing is responsible for the attachment of RNA to the solid support medium.
C	The isolated mRNA in Step 3 do not contain introns.
D	The RNA that attached to the chromatography column lack poly-(A) tails.

9 When a mutation occurs at the gene coding for protein A1, a structurally similar mutant protein, protein A2, is synthesised instead. The two proteins differ by only three amino acids, at positions 14, 16 and 20.

<u>amino acid sequence of normal protein A1</u>



<u>amino acid sequence of mutant protein A2</u>



The table shows the mRNA codons for the amino acids in these positions for proteins A1 and A2.

amino acid	mRNA codons
ala	GCU
ser	AGU UCA
pro	CCA
leu	CUU UUA
lys	AAA
ala	GCU
met	AUG
his	CAU CAC
asn	AAU
val	GUC

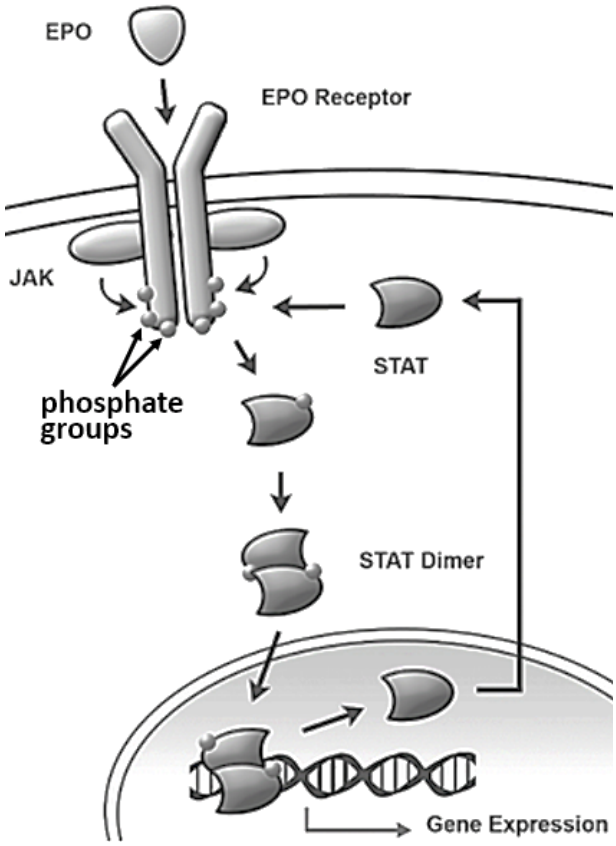
Which combination of mutations occurred in order to account for the amino acid sequence of the mutant protein A2?

1 Frameshift mutation resulting from an addition of a single nucleotide in the codon that codes for amino acid at position 14.

	2 Substitution of a single nucleotide in the codon that codes for amino acid at position 14. 3 Frameshift mutation resulting from a deletion of two nucleotides in the codon that codes for amino acid at position 16. 4 Deletion of a single nucleotide in the codon that codes for amino acid at position 16.

	A	1 and 3
	B	1 and 4
	C	2 and 3
	D	3 and 4

10 The diagram shows the JAK-STAT cell signalling pathway.



Which of the following statements are correct?

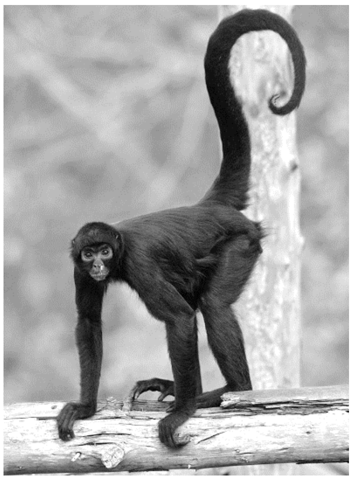
1 EPO is a large and non-polar lipid hormone and hence it needs a cell surface receptor to cross the membrane.

2 Phosphorylation of STAT causes them to dimerize.

3 Gene expression is terminated when phosphatases remove phosphate groups from STAT dimers.

4 Signal amplification occurs as JAK phosphorylates different tyrosine residues on the EPO receptor.

A	1 and 3 only
B	2 and 3 only
C	2 and 4 only
D	2, 3 and 4 only

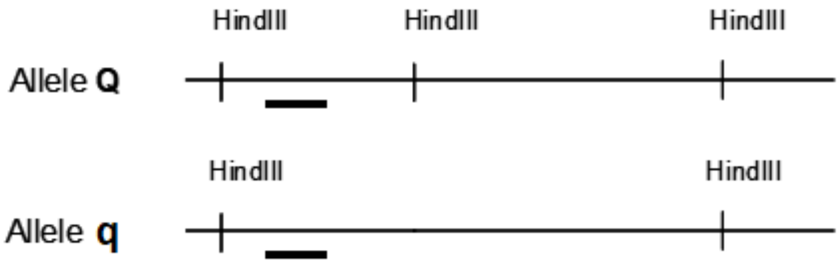
11	Which of the following modification(s) would allow *E. coli* cells to start synthesising a eukaryotic protein? **1** Insertion of the cDNA which was reverse transcribed using the mature mRNA coding for the eukaryotic protein. **2** Presence of activators and histone acetylases. **3** Insertion of a eukaryotic promoter next to the eukaryotic gene in a bacterial plasmid	
	A	**1** only
	B	**1** and **3** only
	C	**2** and **3** only
	D	All of the above.
12	Which of the following is **true** of gene regulation in prokaryotes and eukaryotes?	
	A	Gene regulation in both eukaryotes and prokaryotes can involve proteins binding to DNA.
	B	In eukaryotes, increase in efficiency of translation requires regulatory proteins binding to the mRNA while in prokaryotes, increase in efficiency of translation requires regulatory proteins binding to the ribosomes.
	C	In eukaryotes, RNA polymerase binds to the TATA box while in prokaryotes, RNA polymerase binds to the operator.
	D	Repressors and activators regulate expression in eukaryotes while prokaryotes are regulated by repressors only.
13	Seahorses and spider monkeys both possess a prehensile tail, allowing them to grasp and hold onto structures.   How many of the following statements is/are valid? **1** The prehensile tail is a homologous structure derived from a recent common ancestor shared by the seahorse and spider monkey.	

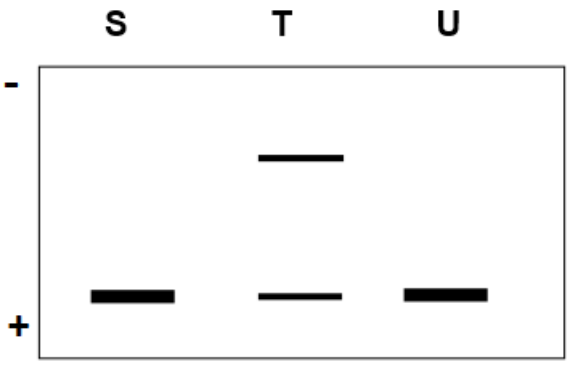
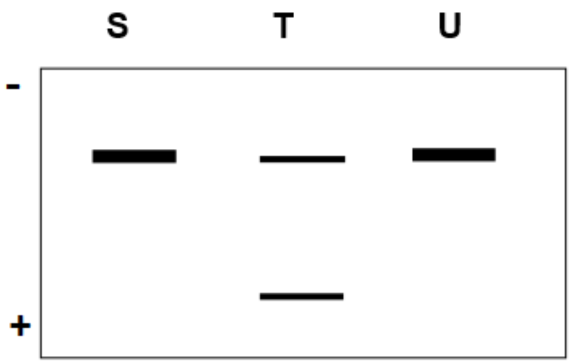
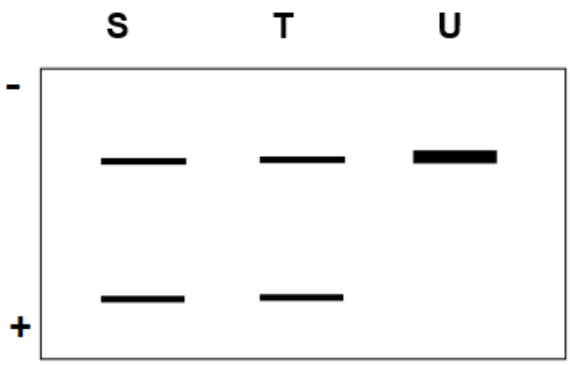
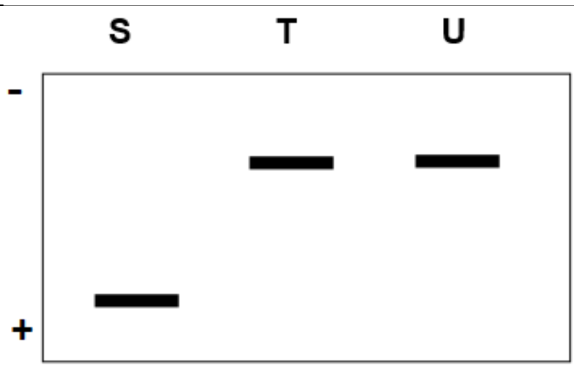
	2 The seahorse and spider monkey evolved the prehensile tail independently due to similar selection pressures in their environment. **3** The prehensile tail is an example of convergent evolution due to both species sharing the same ecological niche. **4** The prehensile tail is an example of descent with modification, as the tail shows modifications in the two species over time.	
	A	1
	B	2
	C	3
	D	All of the above.
14	Relationships between different primates can be found by comparing their proteins and DNA. The protein albumin obtained from a human was injected into a rabbit. The rabbit produced antibodies against the human albumin. These antibodies were extracted from the rabbit and then added to samples of albumin obtained from four different animal species. Precipitation occurs when antibodies bind to albumin. The amount of precipitate produced in each sample was then measured and shown in the table below.	

Species from which albumin was obtained	**Amount of precipitate / arbitrary units**
Rat	80
Chimpanzee	96
Marmoset	85
Trout	45

Which of the following statements is true of the results obtained above?

	A	Rabbit is most closely related to chimpanzee, and least closely related to trout.
	B	The constant region of the rabbit antibodies allows the antibodies to bind to albumin samples from the four animals tested.

	C	Rat and marmoset are more closely related to each other than they are to humans.
	D	Human shares a more recent common ancestor with chimpanzee, followed by marmoset, rat and then trout.
15	Two alleles from a gene located on a particular chromosome is shown below.  *Hind*III indicates the restriction sites for this enzyme, and the black bar indicates the position of the DNA probe used for analysis. DNA fragments from three different individuals, **S**, **T** and **U**, were subjected to restriction digestion by *Hind*III and separated by gel electrophoresis. Given that **S** and **T** are the parents of **U**, which of the following diagrams is NOT a valid result for **U**?	

	A	
	B	
	C	
	D	

16 An analysis of the number of chromosomes in gametes obtained after the meiosis of 2 human germ cells are shown in the table below.

Human germ cell X undergoes meiosis to produce gametes 1-4, while human germ cell Y undergoes meiosis to produce gametes 5-8.

Human germ cell X		**Human germ cell Y**	
Gamete	**Number of chromosomes**	**Gamete**	**Number of chromosomes**
1	22	5	23
2	24	6	23
3	24	7	22
4	22	8	24

Which of the following descriptions is consistent with the data shown in the table above?

A	Homologous chromosomes could not pair up to form bivalents in germ cell X.
B	Non-disjunction of a pair of chromatids during anaphase II in germ cell X.
C	Centromere of a pair of chromatids failed to divide during anaphase II in germ cell Y.
D	Fusion of Gametes 2, 3, and 8 with a normal gamete will always result in offspring with Down Syndrome.

17 Certain types of cancer are associated with chromosomal rearrangements.

Which of the following is a possible consequence when a tumour suppressor gene is translocated within the centromeric sequence of another chromosome?

1 Gene is transcriptionally active.
2 Gene codes for a non-functional tumour suppressor protein.
3 The centrometric sequence is transcribed together with the gene.
4 Cell is unable to carry out DNA repair.

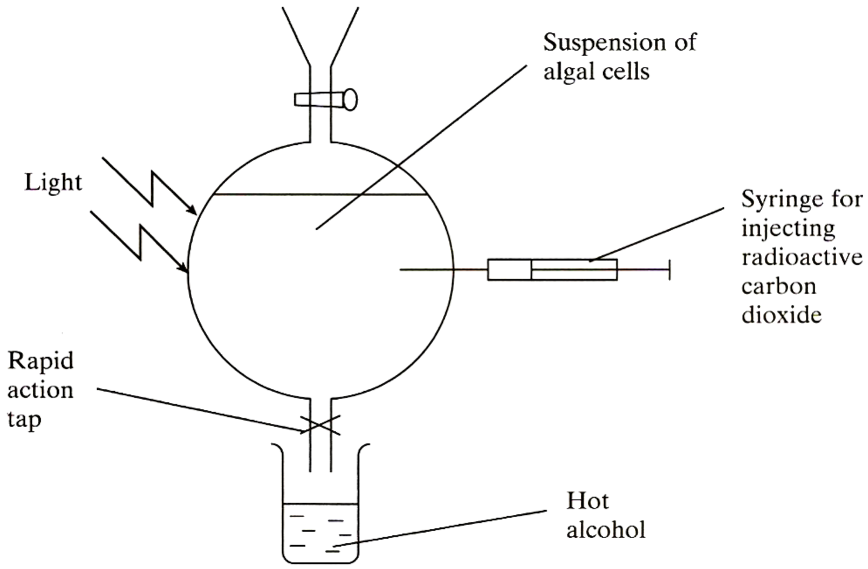
A	**1** and **3**
B	**2** and **3**
C	**2** and **4**
D	**4** only

18	In cats, coat colour is determined by the X-linked, codominant alleles: black and orange. A calico (black, orange and white fur) female, which is the homogametic sex, is bred many times with a black male. Which of the following is a possible outcome of their offspring? **1** All black offspring are female cats. **2** If any orange offspring are obtained, they are male cats. **3** Probability of a female offspring being calico is 0.25. **4** Cross of a black female offspring with an orange male cat will give 1 calico female : 1 black male in the next generation.	
	A	1 and 3
	B	2 and 3
	C	2 and 4
	D	1, 2 and 4
19	In garden peas, the allele **T** codes for terminal position of flowers and is dominant over the allele **t**, which codes for axil position. In an experiment, two garden pea plants, both heterozygous for terminal flowers, produced 77 offspring with flowers at the terminal position and 43 offspring with flowers at the axil position. A χ^2 test was done to determine whether the results of the experiment follow the expected pattern of inheritance. Given that the χ^2 critical value at 1 degree of freedom and a probability of 0.05 is 3.84, which of the following conclusions can be drawn from the experiment and the χ^2 test? $Note: X^2 = \Sigma \frac{(Observed - Expected)^2}{Expected} = \Sigma \frac{(O-E)^2}{E}$	
	A	Since the χ^2 value obtained from the experiment is less than 3.84, flower position in garden peas involves a single pair of segregating alleles at the **T** locus.
	B	Since the χ^2 value obtained from the experiment is less than 3.84, flower position in garden peas does not involve a single pair of segregating alleles at the **T** locus.
	C	Since the χ^2 value obtained from the experiment is greater than 3.84, flower position in garden peas involves a single pair of segregating alleles at the **T** locus.
	D	Since the χ^2 value obtained from the experiment is greater than 3.84, flower position in garden peas does not involve a single pair of segregating alleles at the **T** locus.

20 An investigation was carried out to find out the sequence of biochemical changes that occur during photosynthesis.

Radioactive carbon dioxide was added to a suspension of algal cells, which was placed under light conditions. At intervals, samples of the suspension were removed and dispensed into hot alcohol. These samples were analysed for different radioactively labelled compounds.

The experimental setup is shown below.



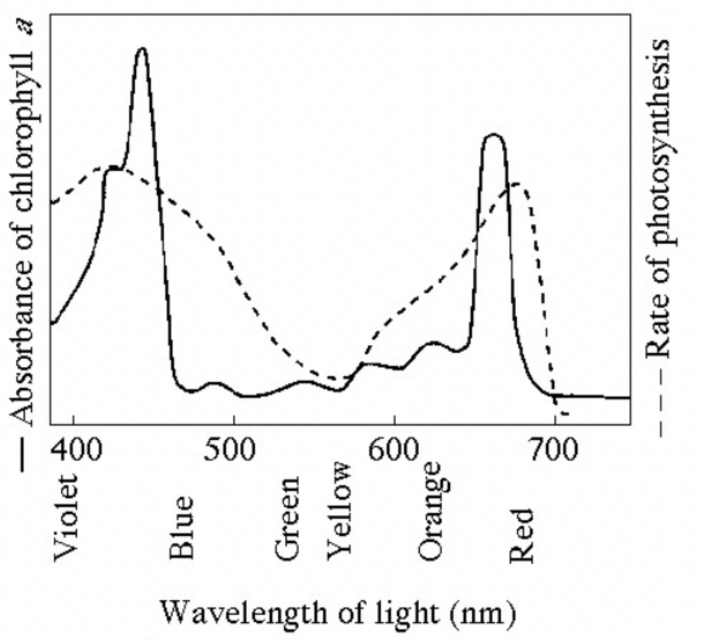
Samples were removed from the suspension at four different times, between 5 seconds and 600 seconds after the start of the experiment. In each sample, the amount of radioactivity present in three different organic compounds, **P**, **Q**, and **R** was measured, and shown in the table below.

organic compound	**amount of radioactivity present / arbitrary units**			
	5 s	60 s	180 s	600 s
P	0.01	0.08	0.17	0.67
Q	1.00	3.10	3.15	3.15
R	0.05	0.16	1.00	1.00

What are the most likely identities of the three organic compounds?

	P	Q	R
A	Rubisco	Glycerate 3-phosphate (GP)	NADP
B	Triose phosphate (TP)	RuBp	Glycerate 3-phosphate (GP)
C	5-carbon compound	Glycerate 3-phosphate (GP)	Triose phosphate (TP)
D	Triose phosphate (TP)	5-carbon compound	6-carbon compound

21 The graph shows the absorption spectrum for chlorophyll *a* and the action spectrum of a plant.



What is the reason for the difference in the absorption spectrum and the action spectrum?

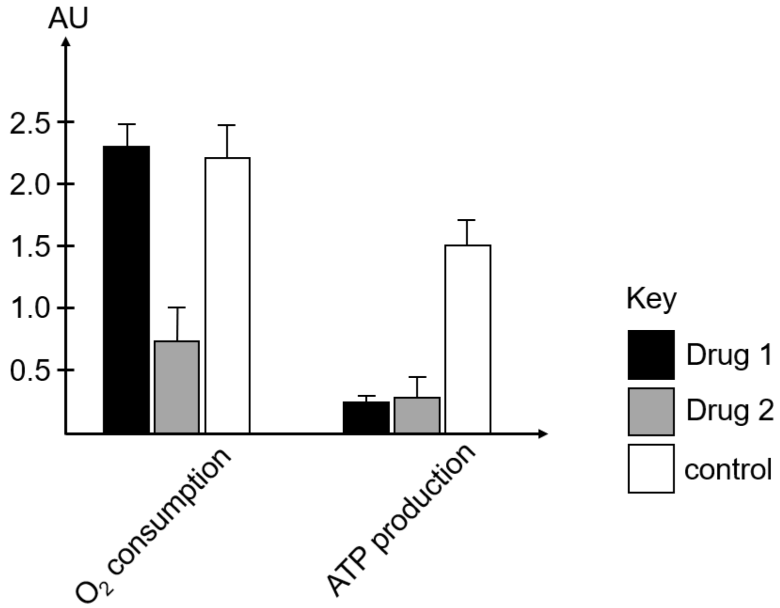
A	Wavelengths of 480-630nm provide the least energy for photosynthesis.
B	Chlorophyll *a* is the main pigment responsible for photosynthesis in the plant.
C	There are other pigments responsible for photosynthesis in the plant.
D	There are other limiting factors affecting the rate of photosynthesis in the plant.

22 Which of the following processes occur during glycolysis:

1 Two 3-carbon compounds combine to form 6-carbon compound
2 Dehydrogenation of substrate
3 Oxidative decarboxylation
4 Substrate-level phosphorylation

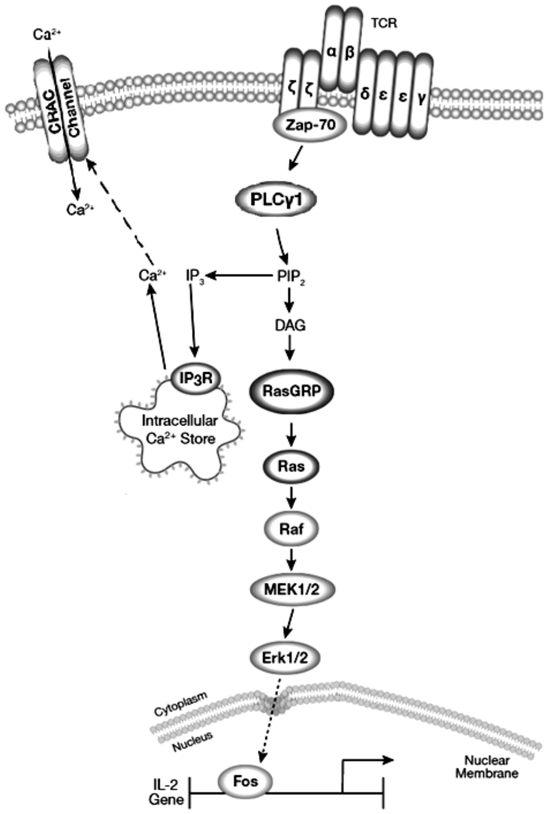
A	1 and 3 only
B	2 and 4 only
C	1, 2 and 4 only
D	2, 3 and 4 only

23 Two drugs, Drug 1 and Drug 2, inhibit different proteins found in mitochondria. Their effects on oxygen consumption and ATP production are recorded in the figure below.



Which of the following correctly identifies possible inhibitory mechanisms of Drugs 1 and 2?

	Drug 1	Drug 2
A	Inhibits ATP synthase	Inhibits a dehydrogenase involved in glycolysis
B	Inhibits cytochrome oxidase	Inhibits an electron carrier
C	Inhibits an enzyme in Krebs cycle	Inhibits ATP synthase
D	Provides an alternative pathway for H^+ to pass through the inner mitochondrial membrane	Inhibits cytochrome oxidase

24	T Cell Receptor (TCR) activation promotes several signalling cascades that ultimately determine cell fate. The diagram below shows part of the cell signalling pathway.  Which of the following can be inferred from the diagram? 1 Zap-70 activates phospholipase C (PLCγ1), which hydrolyses PIP_2 to produce the second messengers diacylglycerol (DAG) and inositol trisphosphate (IP_3). 2 IP_3 triggers the release of Ca^{2+} from the intracellular store, which promotes entry of extracellular Ca^{2+} into cells through CRAC channels. 3 Signal is amplified along the phosphorylation cascade involving kinases such as MEK1/2 and ErK1/2. 4 ErK1/2 activates Fos, which increases the transcription of IL-2 gene, leading to leading to production of antibodies.	
	A	1 only
	B	2 and 3 only
	C	1, 2 and 3 only
	D	All of the above

25 *Greto oto* is a species of clearwing butterfly that has a proportion of its wings having no coloured scales, making them partially transparent.

Different populations of *Greto oto* have varying degree of transparency (i.e. area of wing with no coloured scales), depending on their habitat heterogeneity. Habitat heterogeneity is measured by considering the number and evenness of discreet structural elements in a habitat; the higher the number, the more complex it is.

A researcher glued 30 butterflies (all had wing transparency of 75%) at four different locations and measured the predation rate after 24 hours (defined as the percentage of glued butterflies removed by predators). He also captured 30 live butterflies from the same four locations and recorded their degree of wing transparency.

The table below shows the relationship between habitat heterogeneity, predation rate and degree of wing transparency.

Population	Habitat heterogeneity	Predation rate / %	Degree of wing transparency / %
1	8.2	67.2	72.4
2	6.9	75.4	80.6
3	4.5	83.0	85.3
4	2.1	89.6	90.7

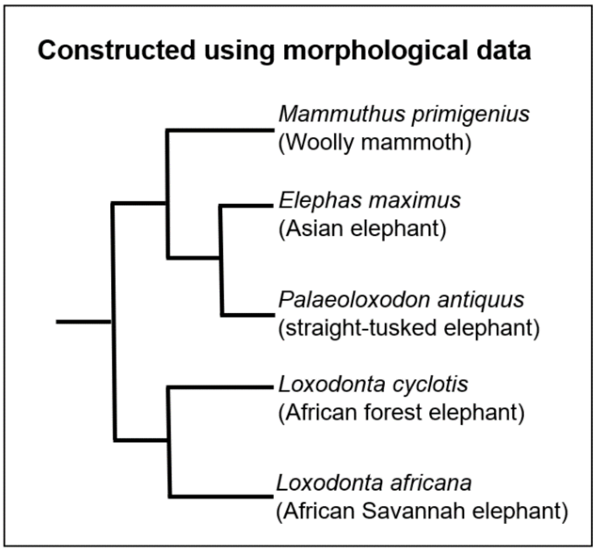
Which of the following can be deduced?

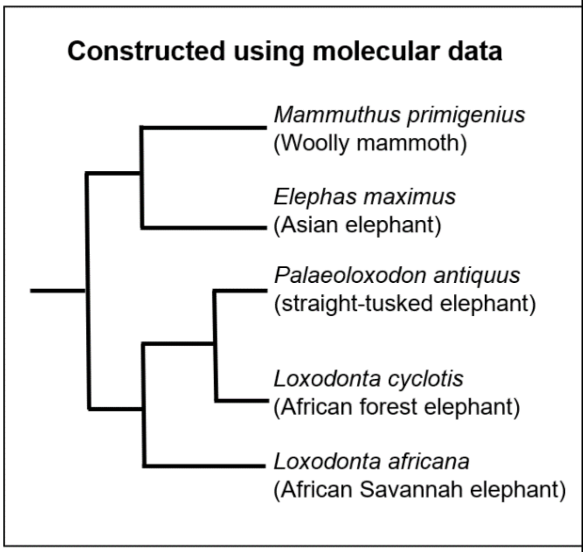
1 Predators have less success in habitats that are more complex.
2 Butterflies with higher degree of wing transparency face higher predation rates.
3 Predation acts as a selection pressure that favors wings that are more transparent.
4 Female butterflies preferred to mate with colourful males (i.e. more coloured scales on wings).

A	1 and 2 only
B	1 and 3 only
C	1 and 4 only
D	3 and 4 only

26 The straight-tusked elephants, *Palaeoloxodon antiquus,* were widespread across Eurasia during the Pleistocene. Phylogenetic reconstructions using morphological traits have grouped them with Asian elephants (*Elephas maximus*). The recovery and analyses of mitochondrial genomes and nuclear genomes from two *Palaeoloxodon antiquus* fossils suggest that *Palaeoloxodon antiquus* was a close relative of extant African forest elephants (*Loxodonta cyclotis*) instead.

The phylogenetic trees below show the evolutionary relationships between the elephants, using morphological and molecular data respectively.

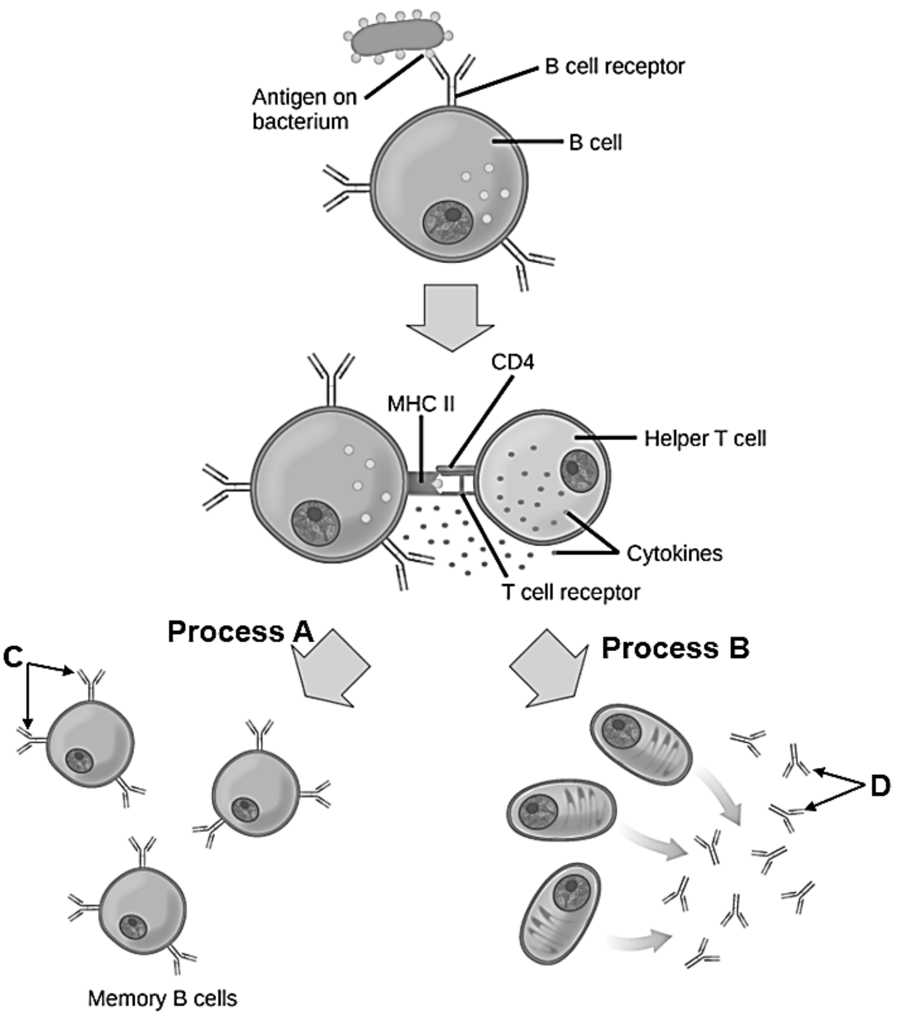


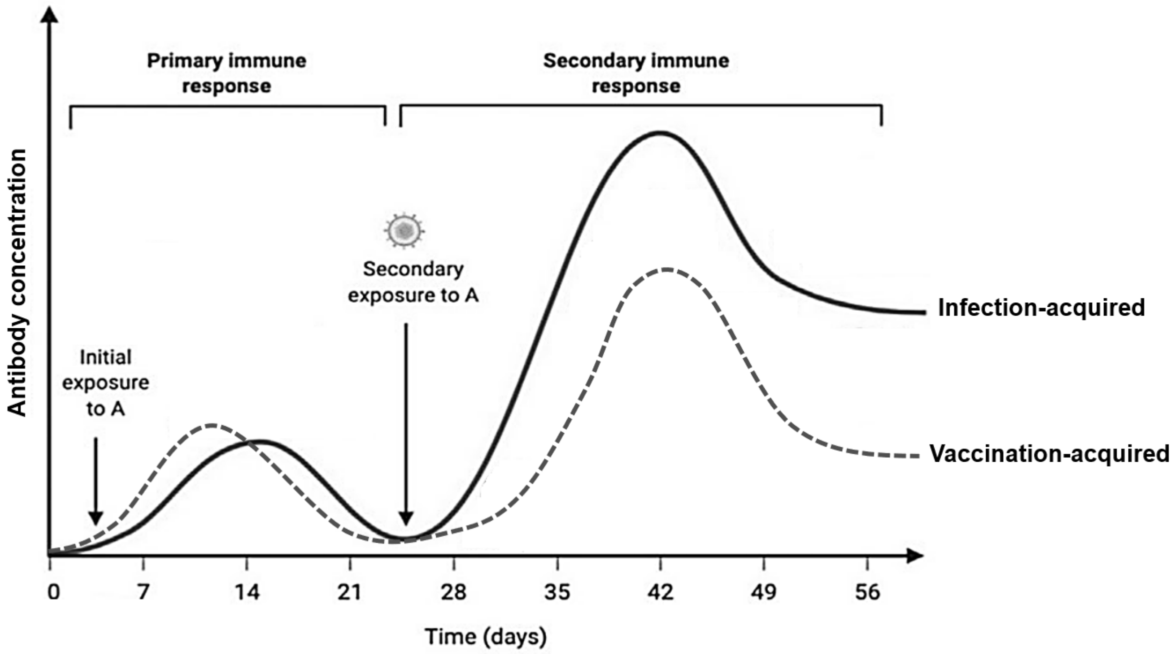
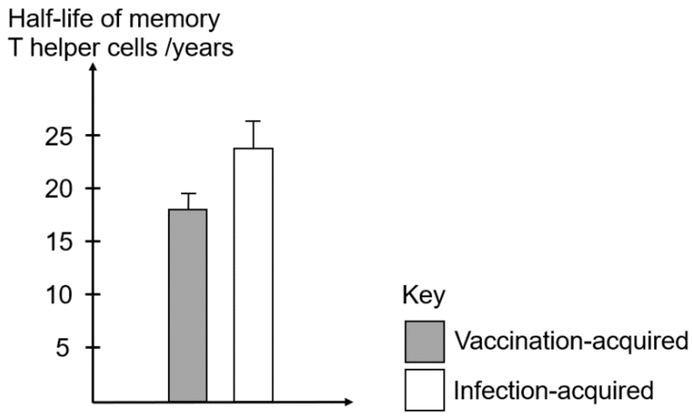


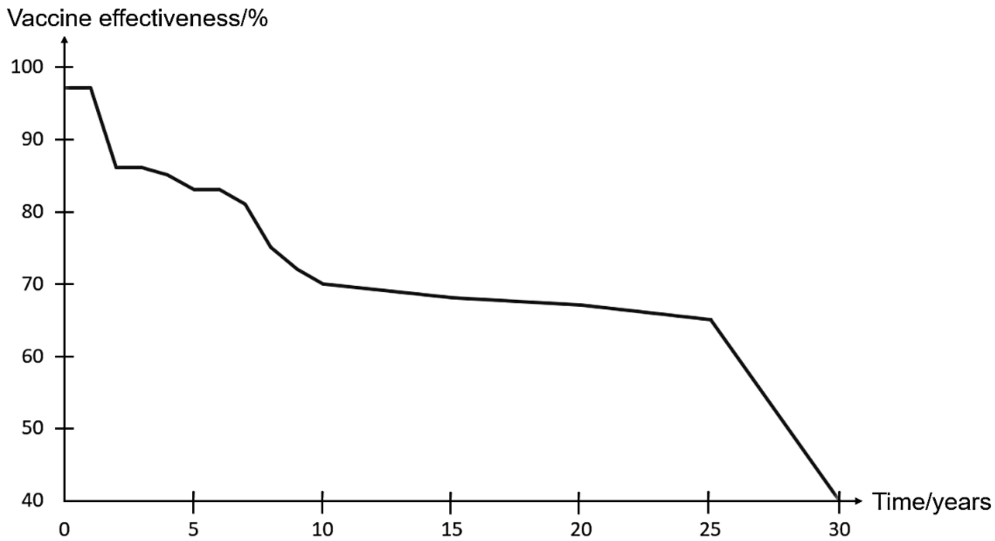
Which of the following statements can be deduced?

1. *Palaeoloxodon antiquus* and *Elephas maximus* share more morphological similarities with each other than they did with *Loxodonta cyclotis*.
2. *Palaeoloxodon antiquus* share more morphological similarities with *Mammuthus primigenius* than it did with *Loxodonta cyclotis*.
3. *Loxodonta cyclotis is more closely related to Loxodonta africana than it is to Palaeoloxodon antiquus because they are the same genus.*
4. *Palaeoloxodon antiquus* is more closely related to *Loxodonta africana* than it is with *Elephas maximus*.

A	1 and 2 only
B	1 and 3 only
C	3 and 4 only
D	1, 2 and 4 only

27	The diagram below shows some processes of the adaptive immune system.  Which of the following are true? 1 Processes **A** produces cells that are involved in the primary response. 2 Process **B** includes somatic recombination and somatic hypermutation. 3 Activated B cells secrete cytokines during Process **B.** 4 Structures **C** and **D** differ in the constant region of the molecule.	
	A	4 only
	B	1 and 3 only
	C	2 and 4 only
	D	2, 3 and 4 only

28	A study was conducted to investigate the difference between adaptive immunity acquired via normal infection vs vaccinations. The researchers obtained two comparative groups of people: one group was hospitalized from being infected with pathogen A and one group was vaccinated against pathogen A. The vaccine contained a mixture of purified viral envelope proteins. They measured the antibody concentration during primary and secondary response, the half-life of the memory T helper cells produced against pathogen A and the effectiveness of the vaccine over a 30-year period. The diagrams below show the results of the study.   *Half-life is defined as time required for number of T helper cells to decrease by half.

29	

Which of the following can be deduced from the results of this study?

1. A vaccinated person produces less antibodies when exposed to the actual pathogen A compared to a recovered patient who is exposed to pathogen A again.
2. If a patient was infected by pathogen A when he was 10 years old, he would have no more memory T cells specific to pathogen A when he turns 60.
3. The infection-acquired immunity will be better than vaccination-acquired immunity to protect against re-infection should pathogen A mutate.
4. The vaccine will no longer be able to protect individuals 30 years after it was given.

A	1 only
B	2 and 3 only
C	1 and 4 only
D	3 and 4 only

29 The table below records data collected from four countries.

Country/Region	Latitude, North/°	Summer temperature range/°C	Life cycle of *Aedes. aegypti* /days	Dengue fever cases per 100,000 people	Projected dengue fever cases per 100,000 people in 2050
Singapore	1.35	26.0 – 32.0	8	577.0	360
Heilongjiang, China	35.9	18.0 – 25.0	21	3.46	50
India	20.6	25.0 – 42.0	10	229.1	120
London, UK	37.1	15.0 – 23.0	No data	0	30

The projected dengue fever cases were calculated based on a 2°C increase in global temperatures predicted by current climate studies.

Which of the following statements are possible explanations of the trends shown in the data?

1. Temperatures are higher at lower latitudes (nearer to the equator) mainly due to increased deforestation, releasing more greenhouse gases. ×

	2 *Aedes aegypti* does not live in regions colder than 23°C. 3 In 2050, regions in higher latitudes are predicted to become more hospitable to *Aedes aegypti* but regions in lower latitudes become less hospitable. 4 *Aedes aegypti* population decreases as duration of life cycle decreases, causing the number of projected dengue fever cases in 2050 to drop. ×
	A 1 and 2 only
	B 2 and 3 only
	C 3 and 4 only
	D 2, 3 and 4 only

30 Which of the following correctly matches the example of human activity to its direct impact?

	Human activity	Impact
1	Increased beef consumption	Increased CO_2 and methane sequestered in their sinks
2	Increased burning of fossil fuel	Increased food insecurity, especially in temperate countries
3	Increased livestock farming	Intensifies water cycle
4	Increased oil palm plantations	Decreased potential for biomedicine discovery

A	3 only
B	1 and 4 only
C	3 and 4 only
D	1, 2 and 3 only

End of Paper